

PRACTICE PROGRAMS – 08.08.2026
BHEESHMA L
192511332

**SIMATS Online Programming Lab**
JAVA PROGRAMMING

Bheeshma L
192511332RC

QUESTIONS 4 / 20 completed

Animal-1

ANIMAL-1
Write a Java program to implement a basic animal classification system using inheritance. Create a parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from the Animal class and have their own unique methods for eating and reproducing.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int primes = 0, composites = 0;
6         while (sc.hasNextInt()) {
7             int n = sc.nextInt();
8
9             if (n > 1) {
10                 boolean isPrime = true;
11                 for (int i = 2; i * i <= n; i++) {
12                     if (n % i == 0) {
13                         isPrime = false;
14                         break;
15                     }
16                 }
17                 if (isPrime) primes++;
18                 else composites++;
19             }
20         }
21         System.out.println("Prime numbers: " + primes);
22     }
23 }
```

OUTPUT

Prime numbers: 4
Composite numbers: 5

INPUT

2 3 4 5 6 7 8 9 10

**SIMATS Online Programming Lab**
JAVA PROGRAMMING

Bheeshma L
192511332RC

PRACTICE PROGRAMS-06/08/26

QUESTIONS

Q1
Euler's
10 marks

Q2
Area
10 marks

Q3
Tax
10 marks

Q4
BCD to
10 marks

4/4 answered

TAX 10 marks

Write a program to calculate tax given following condition :
If income is less than or equal to 150000 then no tax
If taxable income is 150001 – 300000 the charge 10% tax
If taxable income is 300001 – 500000 the charge 20% tax
If taxable income is above 500001 the charge 30% tax

Your Code

```
1 import java.util.Scanner;
2
3 public class TaxCalculator {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter income: ");
8         double income = sc.nextDouble();
9
10        double tax;
11
12        if (income <= 150000) {
13            tax = 0; // no tax "then no tax" (no tax mentioned for <=)
14        } else if (income <= 300000) {
15            tax = 0.10 * income;
16        } else if (income <= 500000) {
17            tax = 0.20 * income;
18        } else {
19            tax = 0.30 * income;
20        }
21
22        System.out.println("Tax = " + tax);
23        sc.close();
24    }
25 }
```

Input

400000

Output

Enter income: Tax = 120000.0

Run Save

PRACTICE PROGRAMS – 08.08.2026
BHEESHMA L
192511332



SIMATS Online Programming Lab
JAVA PROGRAMMING

Bheeshma L

192511332

QUESTIONS

Q1
Student
10 marks

Q2
ATM
10 marks

Q3
TAX
10 marks

Q4
BONUS
10 marks

4/4 answered

Q4 BONUS

10 marks

a bonus 5% salary is given to grade A workers and 10% bonus to Grade B. write a program to enter salary and grade of employee. If salary less than employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get

Your Code

```
import java.util.Scanner;

public class BonusProgram {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double salary;
        char grade;

        System.out.println("Enter salary: ");
        salary = sc.nextDouble();

        System.out.println("Enter grade (A/B): ");
        grade = sc.next().charAt(0);

        double bonusRate = 0.0;
        if (grade == 'A') {
            bonusRate = 0.05;
        } else if (grade == 'B') {
            bonusRate = 0.10;
        } else {
            System.out.println("Invalid grade. Please enter A or B.");
            sc.close();
            return;
        }

        double extraBonusRate = 0.0;
        if (salary < 5) {
            extraBonusRate = 0.02;
        }

        double totalBonus = salary * (bonusRate + extraBonusRate);
        double finalSalary = salary + totalBonus;

        System.out.println("Bonus to be given: " + totalBonus);
        System.out.println("Salary after bonus: " + finalSalary);
        sc.close();
    }
}
```

Input

200000
A

Output

Bonus salary: 200000.0
Bonus to be given: 10000.0
Salary after bonus: 210000.0

Run

Save



SIMATS Online Programming Lab
JAVA PROGRAMMING

Bheeshma L

192511332

PRACTICE PROGRAMS-06/08/26

Back

QUESTIONS

Q1
Student
10 marks

Q2
ATM
10 marks

Q3
TAX
10 marks

Q4
BONUS
10 marks

4/4 answered

Q2 ATM

10 marks

write a program to print the total amount available in the atm machine with the conditions applied total denominations are 2000, 500, 200,100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Your Code

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] denom = {2000, 500, 200, 100};
        int[] counts = new int[4];

        System.out.println("Enter the priority order (as 4 denominations)");
        for (int i = 0; i < 4; i++) {
            denom[i] = sc.nextInt();
        }

        System.out.println("Enter number of notes for each denomination");
        for (int i = 0; i < 4; i++) {
            System.out.print("Notes for " + denom[i] + ": ");
            counts[i] = sc.nextInt();
        }

        int total = 0;
        for (int i = 0; i < 4; i++) {
            total += denom[i] * counts[i];
        }

        System.out.println("Total amount available in ATM: " + total);
        sc.close();
    }
}
```

Input

2000 500 200 100
5 10 2 1

Output

Enter the priority order (as 4 denominations): Enter number of notes for each denomination in the same priority order.
Notes for 2000: Notes for 500: Notes for 200: Notes for 100:

Run

Save